

Euclid's Elements — Book I

Proposition 30

Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

Lines parallel to the same line are parallel to each other

1 Introduction

Proposition I.30 states a structural property of Euclidean parallelism. If AB and CD are both parallel to the same straight line EF , then AB and CD are parallel to one another.

In Euclid's proof the result is obtained indirectly through angle relations: a transversal is introduced, Proposition I.29 is applied twice, angle congruence is transported by transitivity, and Proposition I.27 closes the argument. The current Hilbert reconstruction exposes a different structure. It proves the same statement directly from uniqueness of a parallel through a point, that is, from the Euclidean parallel layer represented by Hilbert's Group IV.

A formal subtlety must be stated explicitly. In the project, `Geo.Parallel` is strict: the two point-lines are nondegenerate, distinct, and disjoint. Consequently the formal theorem contains the additional hypothesis

$$\text{PointLine}(A, B) \neq \text{PointLine}(C, D).$$

In the classical picture this distinctness is implicit in the fact that the three straight lines are drawn as separate lines.

2 Formal Statement and Public Architecture

The current production theorem is

```
import CGJteamLab.HilbertInterface

namespace Geometry

universe u
```

```

variable {Geo : Geometry.Geo.{u}}

theorem euclid_proposition_30
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D E F : Geo.Point)
  (hAB_EF : Geo.Parallel A B E F)
  (hCD_EF : Geo.Parallel C D E F)
  (hDistinct :
    Geo.PointLine A B != Geo.PointLine C D) :
  Geo.Parallel A B C D := by

exact
  hilbert_parallel_transitive_distinct
    Geo
    A B C D E F
    hAB_EF
    hCD_EF
    hDistinct

end Geometry

```

Thus `euclid_proposition_30` is a thin wrapper. It contains no proposition-local helper theorem, no proposition-local geometric construction, and no case split. Its direct dependency is

$$\text{euclid_proposition_30} \longrightarrow \text{hilbert_parallel_transitive_distinct}.$$

The public typeclasses are

```

[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]

```

so, unlike I.31, Proposition I.30 lives explicitly in the Euclidean parallel layer.

3 Classical Configuration

Let AB , CD , and EF be three distinct straight lines with

$$AB \parallel EF, \quad CD \parallel EF.$$

Euclid introduces one transversal meeting the three lines at G , H , and K . This is the first genuinely new geometric state in the classical proof.

The two applications of I.29 give the chain

$$\angle AGK \cong \angle GHF, \quad \angle GHF \cong \angle GKD.$$

By transitivity,

$$\angle AGK \cong \angle GKD.$$

These are alternate interior angles for the first and third lines, so I.27 gives

$$AB \parallel CD.$$

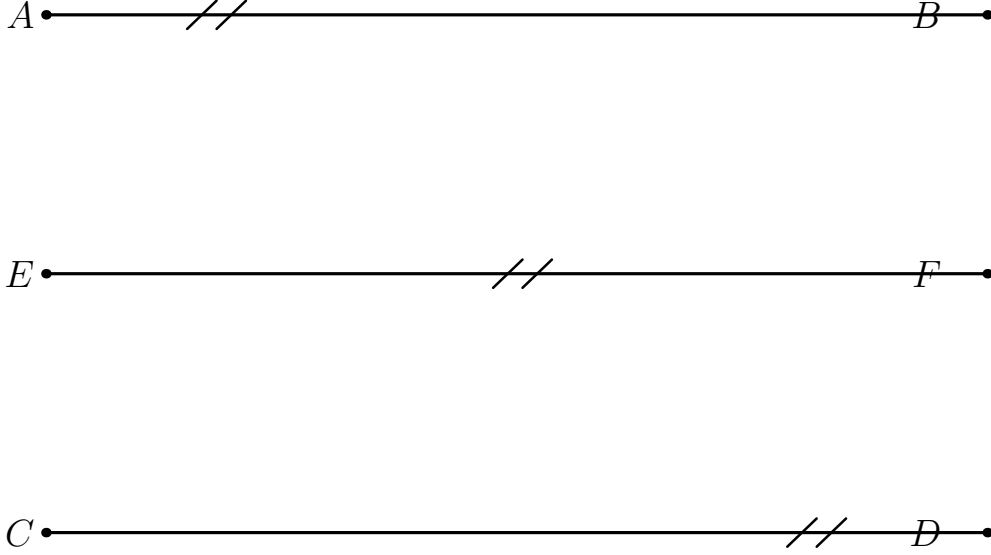


Figure 1: The classical configuration for Proposition I.30. The two given parallel relations are encoded separately: $AB \parallel EF$ and $CD \parallel EF$. Euclid introduces the dashed transversal GKH . The diagram deliberately does not mark $AB \parallel CD$, since that is the conclusion of the proposition.

The classical dependency graph is therefore

$$\begin{array}{ccc}
 AB \parallel EF & & CD \parallel EF \\
 \downarrow \text{I.29} & & \downarrow \text{I.29} \\
 \angle AGK \cong \angle GHF & & \angle GHF \cong \angle GKD \\
 & \swarrow \quad \downarrow \quad \searrow & \\
 & \angle AGK \cong \angle GKD & \\
 & \downarrow \text{I.27} & \\
 & AB \parallel CD. &
 \end{array}$$

Thus the literal Euclidean route is angle-theoretic even though the theorem's structural content is a transitivity property of parallelism.

4 Hilbert Reconstruction

The reusable theorem is

```

theorem hilbert_parallel_transitive_distinct
  [HilbertEuclideanPlane Geo]
  (A B C D E F : Geo.Point)
  (hAB_EF : Geo.Parallel A B E F)
  (hCD_EF : Geo.Parallel C D E F)
  (hDistinct :
    Geo.PointLine A B != Geo.PointLine C D) :
  Geo.Parallel A B C D

```

The proof first chooses incidence carriers

$$\ell_{AB}, \quad \ell_{CD}, \quad \ell_{EF}$$

through the three endpoint pairs. The two parallel hypotheses are converted into disjointness

statements

$$\ell_{AB} \cap \ell_{EF} = \emptyset, \quad \ell_{CD} \cap \ell_{EF} = \emptyset.$$

To prove $AB \parallel CD$, the proof unfolds the disjointness of the first and third project point-lines. It assumes a hypothetical common point P :

$$P \in \text{PointLine}(A, B), \quad P \in \text{PointLine}(C, D).$$

Membership is transported to incidence on the selected Hilbert lines, so $P \in \ell_{AB}$ and $P \in \ell_{CD}$. Since ℓ_{AB} is disjoint from ℓ_{EF} , one has

$$P \notin \ell_{EF}.$$

Now both ℓ_{AB} and ℓ_{CD} pass through the same point P outside ℓ_{EF} , and each is disjoint from ℓ_{EF} . The call

`HilbertEuclideanPlane.parallel_unique`

therefore gives

$$\ell_{AB} = \ell_{CD}.$$

The equality of Hilbert incidence carriers is then converted back to the project-level extensional carriers by

`hilbert_pointLine_eq_of_points_on_line`,

yielding

$$\text{PointLine}(A, B) = \text{PointLine}(C, D),$$

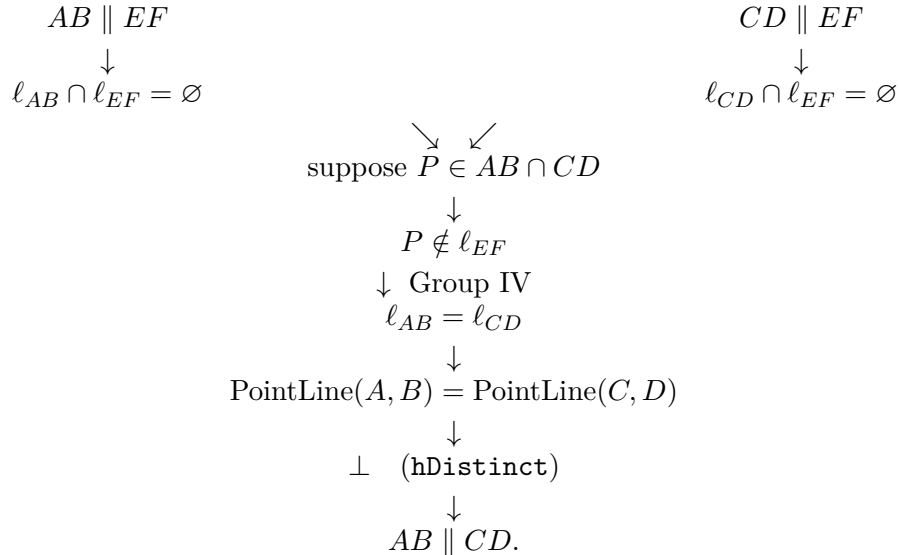
contrary to the explicit hypothesis `hDistinct`.

Hence the assumed common point cannot exist and the two project point-lines are disjoint. This is exactly the desired strict parallelism.

There is no geometric case split. The point P appears only under the contradictory assumption that the first and third point-lines meet.

5 Actual Lean Dependency Graph

The current formal architecture can be summarized as



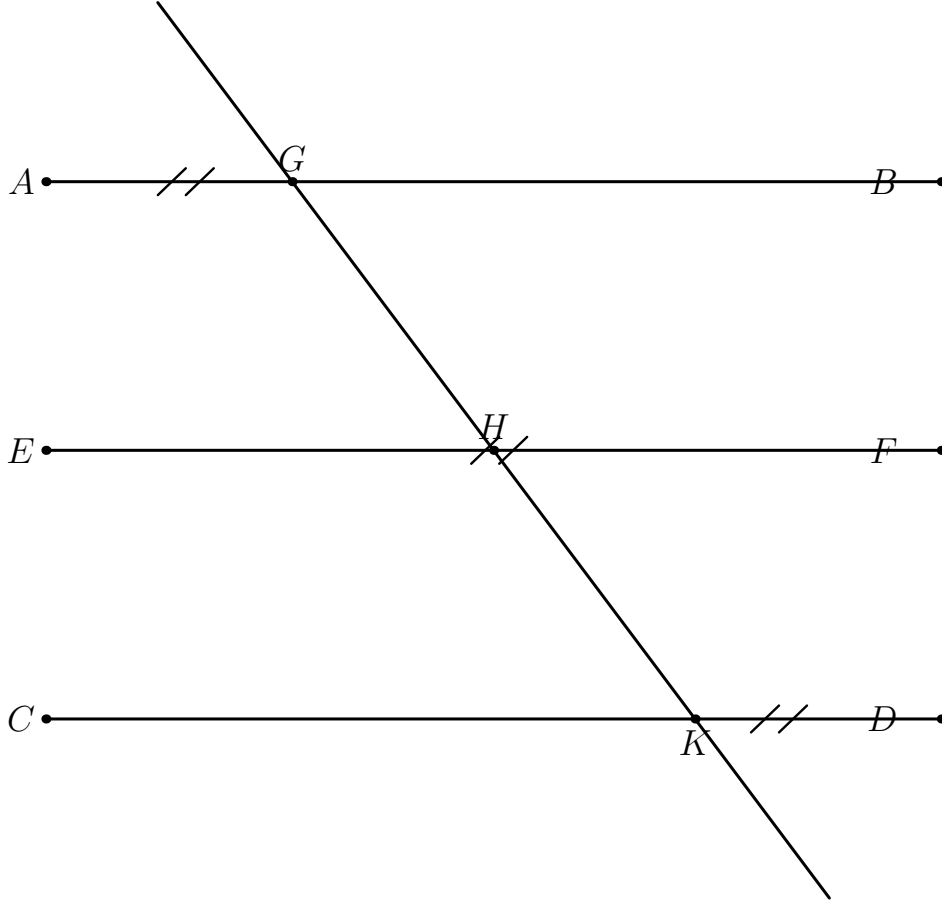


Figure 2: The hidden contradiction state in the current Hilbert proof. One supposes that the two distinct outer point-lines have a common point P , while the hypotheses still assert $AB \parallel EF$ and $CD \parallel EF$. This cannot be realized as a genuine Euclidean configuration; the diagram records only the counterfactual intersection witness. Group IV identifies the two lines through P that are disjoint from EF , contradicting $\mathbf{hDistinct}$.

The formal proof therefore does not call I.29 or I.27. Those propositions are part of Euclid’s historical route, whereas the reusable Hilbert theorem exposes the uniqueness principle directly.

6 Diagrammatic Audit

The preserved Book1R chapter contained two separate diagrammatic ideas, but they were not correctly matched to their captions.

The old initial-data figure showed only the three horizontal lines. As a standalone state it was redundant: the classical proof immediately introduces a transversal, and the richer transversal picture already contains the initial geometry. Moreover, using one identical parallel mark on all three lines visually suggested the conclusion $AB \parallel CD$ before it had been proved.

The preserved Figure 30.2 had a more serious mismatch. Its caption described the Hilbert contradiction argument with a common point P , while the picture actually showed the classical transversal through G, H, K . The caption and the geometry therefore belonged to different proof architectures.

The audited chapter uses exactly two figures:

1. a classical Euclid figure with the transversal GHK , with the two input parallel relations marked separately and without marking the conclusion;
2. a formal counterfactual figure showing the hidden intersection witness P used in the proof by contradiction.

No separate final figure is added for $AB \parallel CD$. After the contradiction no new point, line, intersection, side relation, or construction is introduced; only the logical status of the existing two point-lines changes. A figure should record geometry, not tactic state.

7 Logical Position of Proposition I.30

The sequence I.27–I.31 makes the logical boundary in the parallel theory particularly clear:

- I.27 : equal alternate angles $\implies$ parallel,
- I.28 : other angle conditions $\implies$ parallel,
- I.29 : parallel $\implies$ angle relations,
- I.30 : parallel to the same line $\implies$ parallel to one another,
- I.31 : constructive existence of a parallel.

The first two implications are neutral. I.29 crosses into the Euclidean layer because the reverse angle transport requires uniqueness of parallels. I.30 is another direct manifestation of that same principle. Its current formal proof uses `parallel_unique` exactly at the point where a hypothetical intersection would create two different parallels through one point to the same given line.

In this sense the current reconstruction makes the structural content of I.30 more explicit than the classical angle proof: transitivity for distinct parallels is a consequence of uniqueness.

8 Conclusion

Proposition I.30 has two genuinely different proof pictures. Euclid introduces a transversal and proves the result through I.29, angle transitivity, and I.27. The current Lean development instead assumes a common point of the two outer lines and uses Hilbert’s Euclidean parallel uniqueness to force the carriers to coincide, contradicting their explicit distinctness.

The audited documentation therefore keeps the two architectures separate. It uses one classical transversal figure and one formal contradiction figure, and it removes the former mismatch in which a Hilbert caption was attached to a classical diagram.